

# Outcome Based Assessment

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Score 120

# Assessment is...

a way of finding out what a person knows, understands and thinks, and what he or she can do.

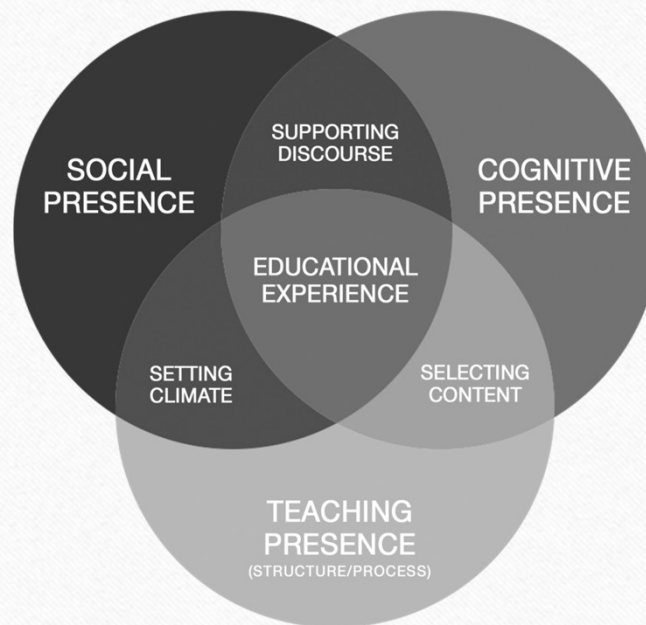
**the systematic collection and analysis of information to monitor and improve student learning.**

is not a one-time event, but a dynamic on-going process.



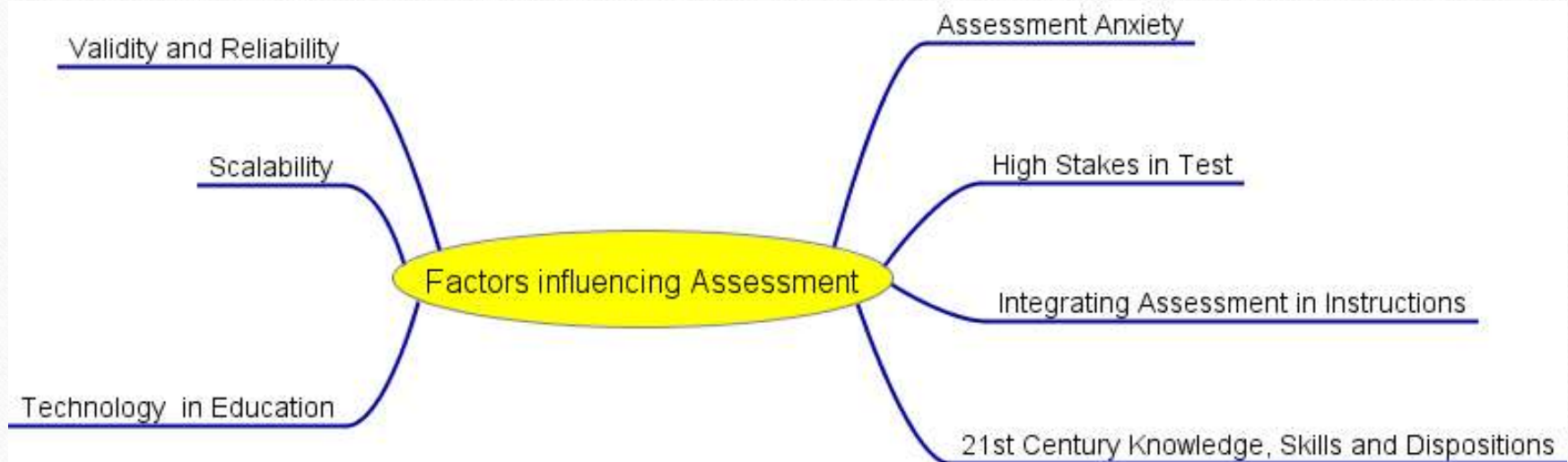
# Good online teachers have regular interaction with their students

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<https://humanmooc.pressbooks.com/>

# Factors influencing Assessment





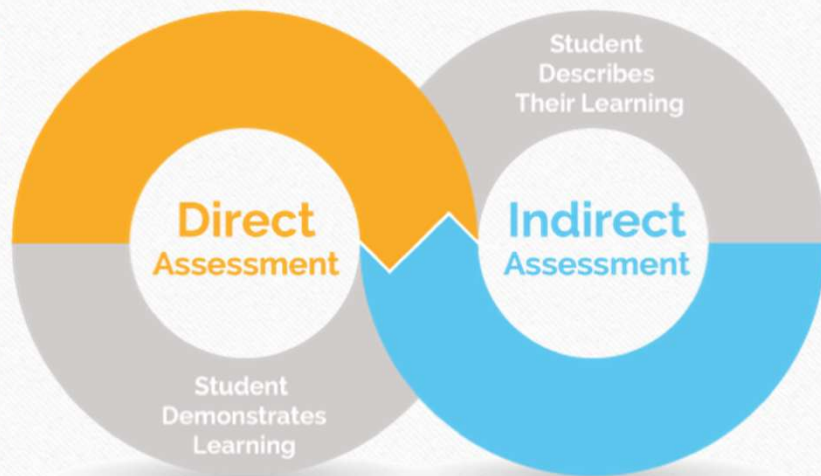
# Integrating Assessment in Instructions - Augmenting Assessment with LMS

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- Learning Analytics
- Individualized Learning and Feedback
- High Quality Assessment
- Peer – to – Peer Interaction
- Tracking the learning and its progress

# Types of Assessment

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## **Direct Assessment**

Ability developed after the learning process ( In the domain of Cognitive, Affective and Psychomotor)

## **Indirect Assessment**

Learning Experience



# Direct Assessment

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- Course-embedded assessment (e.g., homework assignment; essays, locally developed tests)
- Standardized test
- Portfolio Assessment
- Case studies
- Class project (individual or group)
- Internship evaluation
- Thesis evaluation
- Project Report Evaluation

# Indirect Assessment

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- Indirect assessment methods require that faculty infer actual student abilities, knowledge, and values rather than observe direct evidence.



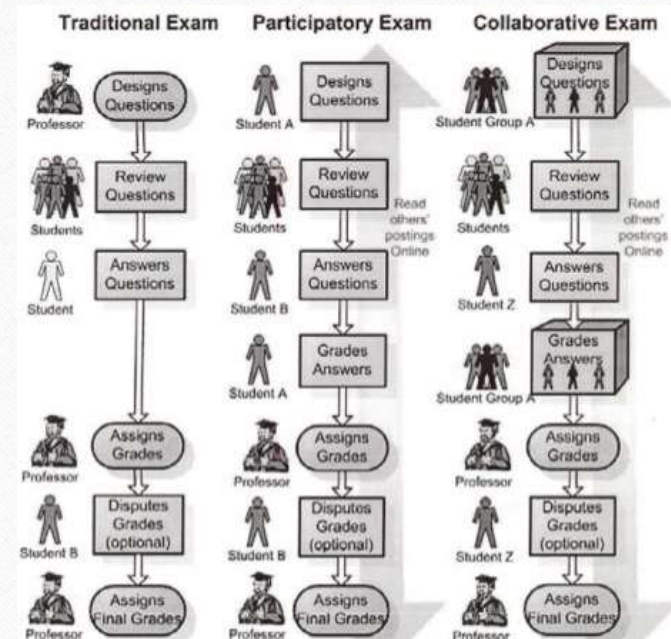
# Indirect Assessment

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- Surveys: Surveys usually are given to large numbers of possible respondents, usually in writing, and often at a distance.
- Exit interviews and focus groups: Exit interviews and focus groups allow faculty to ask specific questions face-to-face with students.
- External reviewers: External reviewers are usually representatives of the discipline and usually are guided by discipline-based standards.

# Methods of Different Assessment strategies

- Traditional Method of Assessment
- Participatory Method of Assessment
- Collaborative Method





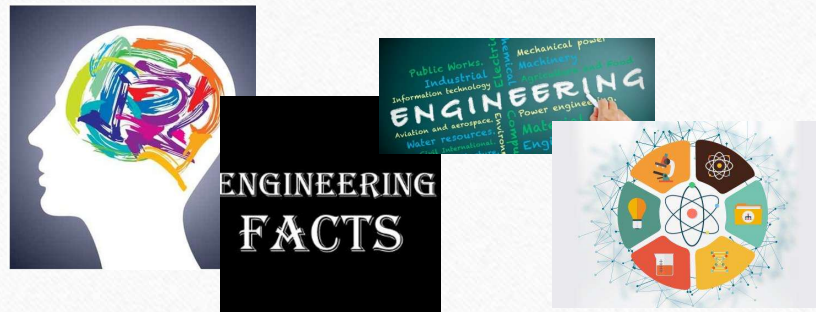
# Assessment Plan

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1. Preparation of Learning Outcomes
2. Assessment plan
  1. Assessment Method
  2. Target Outcomes
  3. Collection of Information and subsequent actions

# Question Paper Preparation

- Question consists of two components  
= Ability Expected + On content

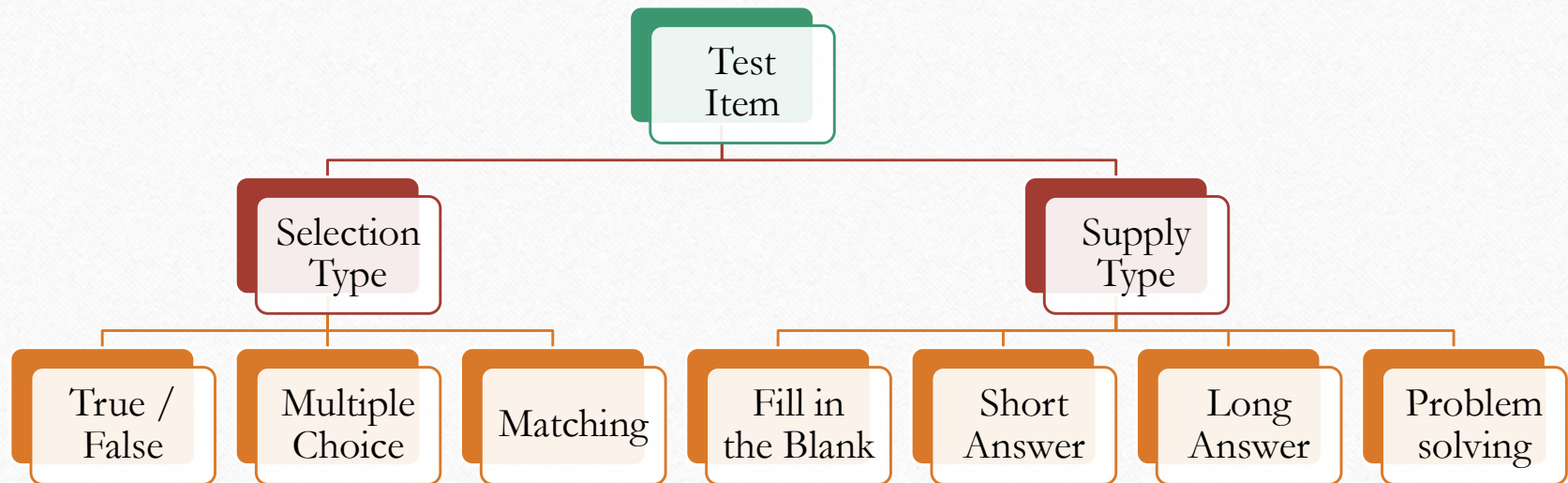




# Assessment Plan

| Assessment Method | Target Outcomes | Subsequent Actions – Feedback |
|-------------------|-----------------|-------------------------------|
| Quiz              | 1,2             |                               |
| Standardized Test | 3,4,5           |                               |
| Lab Assessment    | 5,6,7           |                               |
| Capstone Project  | 8,9             |                               |

# Question Items





## GUIDELINES FOR CONSTRUCTING STRUCTURED ESSAY QUESTIONS

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1. Choose and decide on a structured essay type question only when it could not be tested by short answer type tests.
2. Consider carefully the content and behavior before deciding on the wording of the question.
3. Before the item is considered complete, determine how it is to be scored.

# GUIDELINES FOR CONSTRUCTING STRUCTURED ESSAY QUESTIONS

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4. Structuring of a question is done by breaking the open ended aspect of an essay question by means of a series of tasks to be performed and making them clear, unambiguous and specific. The questions must be interpreted in the same way by the teachers and the students.
5. Relate the questions as directly as possible to the learning outcomes being measured and the question must be a valid testing situation for the ability considered.



# GUIDELINES FOR CONSTRUCTING STRUCTURED ESSAY QUESTIONS

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6. In a set of structured questions, each part should be independent and complete in itself and the correct answer to one part should not rest too much on the correct answer to a former part. At the same time, each of the parts should refer to the same situation and contribute to the overall purposes of the test.
7. The subdivisions in a set of structured questions should be arranged in order of increasing difficulty.

# GUIDELINES FOR CONSTRUCTING STRUCTURED ESSAY QUESTIONS

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8. The relative importance of each subdivision of the questions should be clearly indicated in terms of the marks allotted.
9. When the statement of the question itself does not impose fairly well-defined limits, some indication of the desired length of the answer should be given (Example : Explain within 200 words, write a paragraph, explain in about 5 sentences, not exceeding 1 page etc.) indicating the time expected for answering and the marks.



# GUIDELINES FOR CONSTRUCTING STRUCTURED ESSAY QUESTIONS

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- 10 The directions given in the question should be clear. (Example : Using a diagram, stating the assumptions, when temperature is constant etc).
- 11 Avoid phrases like 'Discuss briefly' 'Explain briefly' etc. as these terms do not specify the exact limit of the answer expected and may be understood differently by different students.

# Suggested terms and situations for essay tests

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- Comparison of two things on a single designed basis, contrast, distinguish, discriminate, differentiate.
- Comparison of two things in general.
- Relationship involving cause and effect.
- Explanation of the use.
- Summary or inference from known principles and facts.



# Suggested terms and situations for essay tests

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- Analysis
- Illustrations(with own examples)
- Classification.
- Application of rules or principles in new situations
- Discussion by multiple interpretation.
- Statement of aim, purpose in his selection, or organization of materials

## Suggested terms and situations for essay tests

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- Criticism on correctness or relevance.
- Outline methods and procedure.
- Explain or define a problem or principle.
- Detailing observations and offering remarks



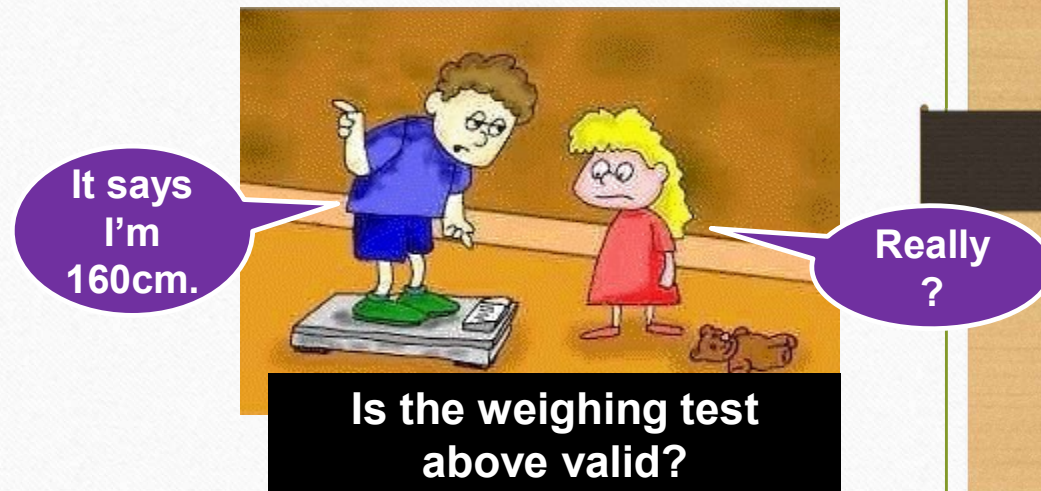
# Characteristics of Assessment

- Validity
- Reliability
- Usability



# Validity

- The term validity refers to whether a test measures what it intends to measure.
- On a test with high validity the items will be closely linked to the test's intended focus





# Table of Specifications

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- The type of things the student should be able to do (i.e. ABILITIES)
- The subject matter in which he should be able to do them (i.e. CONTENT)

A Table of Specifications is a two way-chart, which relates CONTENT and ABILITIES by assigning suitable weightages for testing purpose.

# TABLE OF SPECIFICATIONS

Programme : B. Tech – Information Technology

Semester : V Semester

Subject : Computer communication and Networks (Code Number : XXXXX)

| Content<br>(No. and Name of the Unit) | Remember  |        | Understand |     |     |     |     |     |     | Apply |     | Higher order abilities | Total      |
|---------------------------------------|-----------|--------|------------|-----|-----|-----|-----|-----|-----|-------|-----|------------------------|------------|
|                                       | Recognize | Recall | 2.1        | 2.2 | 2.3 | 2.4 | 2.5 | 2.6 | 2.7 | 3.1   | 3.2 |                        |            |
|                                       |           |        |            |     |     |     |     |     |     |       |     |                        |            |
|                                       |           |        |            |     |     |     |     |     |     |       |     |                        |            |
|                                       |           |        |            |     |     |     |     |     |     |       |     |                        |            |
|                                       |           |        |            |     |     |     |     |     |     |       |     |                        |            |
|                                       |           |        |            |     |     |     |     |     |     |       |     |                        |            |
| <b>Total</b>                          |           |        |            |     |     |     |     |     |     |       |     |                        | <b>100</b> |



# Preparing a Table Of Specifications

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Assigning weightage to the various units of the content

Factors to be considered:

- I.** Number of periods allotted for teaching the unit
- II.** Usefulness of the content matter of the unit in the student's job or every day life
- III.** Usefulness of the content matter of the unit in understanding other units of the same subject
- IV.** Usefulness of the content matter of the unit in understanding other subjects prescribed for the programme

# Subject: Computer communication and Networks

## Weightage for Various Abilities to be tested

| CATEGORY OF ABILITY    | WEIGHTAGE |
|------------------------|-----------|
| Remember (R)           | 20%       |
| Understand (U)         | 60%       |
| Apply(Ap)              | 20%       |
| Higher order abilities | 0%        |
| Total                  | 100%      |



# Content Achievement

| Unit / Module No | Unit Name                                            | Imparting Content in Percentage |
|------------------|------------------------------------------------------|---------------------------------|
| Unit / Module 1  | Network Design                                       | 18%                             |
| Unit / Module 2  | LAN Access methods and Standards                     | 15%                             |
| Unit / Module 3  | Packet Switching Networks                            | 18%                             |
| Unit / Module 4  | TCP / IP Architecture                                | 22%                             |
| Unit / Module 5  | Advanced Network Architecture and Security Protocols | 27%                             |

# TABLE OF SPECIFICATIONS

| Abilities<br>Content<br>(No. and Name<br>of the Unit)      | Remember  |        | Understand |     |     |     |     |     |     | Apply    |          | Higher<br>order<br>abilities | Total      |
|------------------------------------------------------------|-----------|--------|------------|-----|-----|-----|-----|-----|-----|----------|----------|------------------------------|------------|
|                                                            | Recognize | Recall | 2.1        | 2.2 | 2.3 | 2.4 | 2.5 | 2.6 | 2.7 | 3.1      | 3.2      |                              |            |
| Network Design                                             | 5         | 5      | 2          | 0   | 0   | 0   | 0   | 2   | 4   | 3.1      | 3.2      | 0                            | 18         |
| LAN Access methods<br>and Standards                        | 0         | 3      | 0          | 0   | 0   | 4   | 4   | 0   | 4   | 0        | 0        | 0                            | 15         |
| Packet Switching<br>Networks                               | 0         | 3      | 0          | 0   | 0   | 0   | 2   | 4   | 4   | 0        | 0        | 0                            | 18         |
| TCP / IP Architecture                                      | 0         | 2      | 0          | 0   | 0   | 0   | 0   | 3   | 12  | 5        | 0        | 0                            | 22         |
| Advanced Network<br>Architecture and Security<br>Protocols | 0         | 2      | 0          | 0   | 0   | 0   | 0   | 5   | 10  | 5        | 0        | 0                            | 27         |
| <b>Total</b>                                               | <b>20</b> |        | <b>60</b>  |     |     |     |     |     |     | <b>5</b> | <b>5</b> | <b>0</b>                     | <b>100</b> |



# Reliability

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- Reliability refers to the extent to which an assessment method or instrument measures consistently the performance of the student.
- Consistency across time
- Consistency across tasks
- Consistency across markers

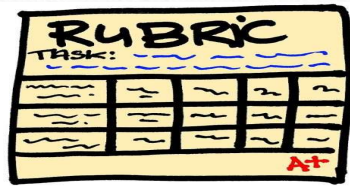
# Practicality

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- Usability of a test includes such practical features as:
  - Ease of administration
  - Time required
  - Ease of scoring
  - Ease of interpretation and cost



# Performance assessment



How to develop a rubric  
for your curriculum

6 steps to create a rubric for classrooms

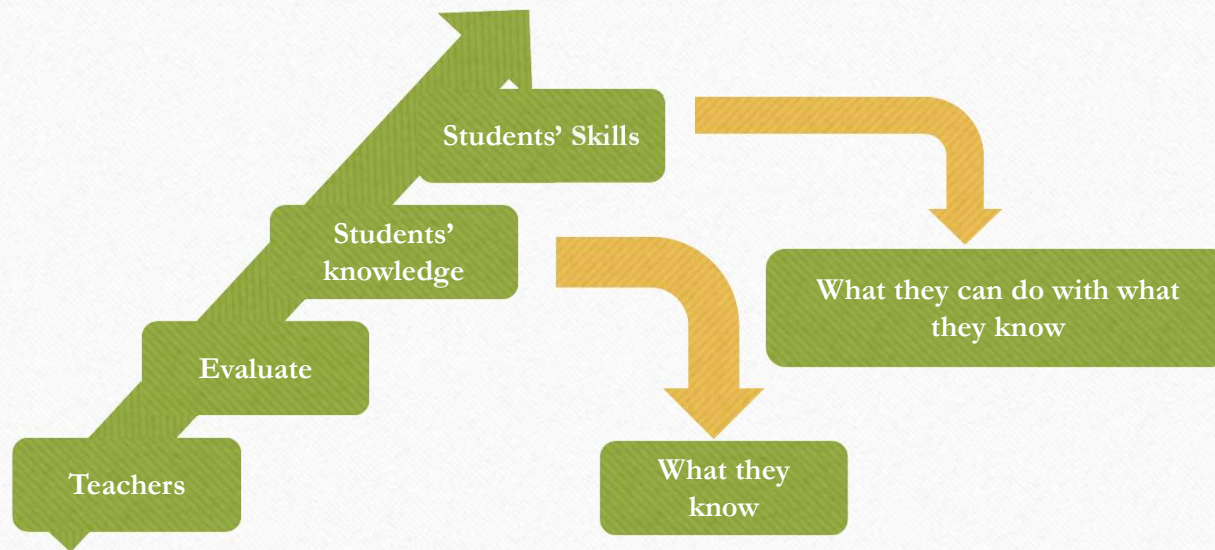


## Portfolio



# Performance Assessment

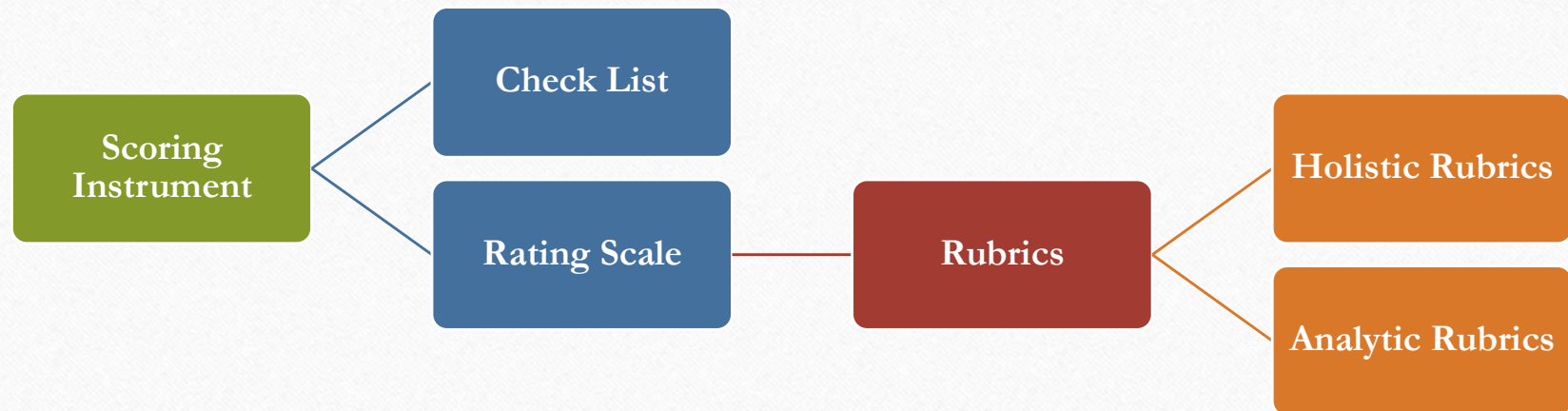
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# Scoring Instruments

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# Rubric

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- A rubric is an assessment tool often shaped like a matrix, which describes levels of achievement in a specific area of performance, understanding, or behavior.



# Components of Rubrics

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- **A task description.** The outcome being assessed or instructions students received for an assignment
- **Criteria**
  - The characteristics to be rated
- **Rating Scale**
  - Levels of mastery/scale
- **Descriptors**
  - A description of each characteristic at each level of mastery/scale

# Holistic Rubric

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- A holistic rubrics provide a single score based on an overall impression of a student's performance on a task.
- **Advantages:** quick scoring; provides an overview of student achievement; efficient for large group scoring
- **Limitations:** does not provided detailed information; not diagnostic; may be difficult for scorers to decide on one overall score
- Use when:
  - You want a quick snapshot of achievement.
  - A single dimension is adequate to define quality.



# Holistic Rubric – Group Participation

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- “Participating” is a matter of active engagement, rather than passive observation, and it is shown through working effectively in diverse groups and teams, as well as through cooperation and respect for others.
- Participation quality will be evaluated using the features defining the four levels in the following example

# Structure of Holistic Rubric

| Rating Scale (Score) | Description |
|----------------------|-------------|
|                      |             |
|                      |             |
|                      |             |
|                      |             |
|                      |             |
|                      |             |



# Analytic Rubric

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| Criteria | Level of Performance |           |      |           |                 |
|----------|----------------------|-----------|------|-----------|-----------------|
|          | Excellent            | Very Good | Good | Satisfied | Need to Improve |
|          |                      |           |      |           |                 |
|          |                      |           |      |           |                 |
|          |                      |           |      |           |                 |

# Portfolio



- A portfolio is a collection of student work that can exhibit a student's efforts, progress, and achievements in various areas of the curriculum.
- Portfolios show the cumulative efforts and learning of a particular student over time. They offer valuable data about student improvement and skill mastery.
- Along with student reflection, that data provides valuable information about how each student learns and what is important to him or her in the learning process.